EduOS Atlas Mock Examination

Pearson Edexcel International GCSE
Biology

Paper 1: Core Biology

Paper Reference **4BI1/1C**

Mock Set 1

Candidate Surname	
Other Names	
Candidate Number	
Centre Number	

Time allowed: 2 hours **Total marks:** 110

Instructions

- Answer ALL questions.
- Write your answers in the spaces provided.
- Use black ink or black ball-point pen.
- Show all working in calculations.
- Calculators may be used.

Information

- The total mark for this paper is 110.
- The marks for each question are shown in brackets.
- Quality of written communication will be assessed in questions marked with an asterisk (*).

Advice

- Read each question carefully before you start to answer it.
- Keep an eye on the time.
- Try to answer every question.

Section A: Multiple Choice (10 marks — 1 mark each)

Answer ALL questions in this section. Circle the letter next to the best answer.

1 Which of the following correctly describes the function of the cell membrane?

A It provides structural support to the cell
B It controls what enters and leaves the cell
C It is the site of protein synthesis
D It stores the cell's genetic information

Circle your answer: A B C D

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2 A student observes a plant cell under a microscope. Which structure would NOT be found in an animal cell?

A Mitochondria
B Cell membrane
C Chloroplast
D Cytoplasm

Circle your answer: A B C D

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3 The equation for aerobic respiration is: glucose + oxygen → carbon dioxide + water + energy
Which organ system is primarily responsible for supplying oxygen to respiring cells?

A Digestive system
B Excretory system
C Circulatory and respiratory systems
D Nervous system

Circle your answer: A B C D

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4 Which enzyme is responsible for breaking down starch in the mouth?

A Lipase
B Protease
C Amylase
D Pepsin

Circle your answer: A B C D

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5 During osmosis, water molecules move:

A From a region of low water potential to high water potential
B From a region of high water potential to low water potential
C Against the concentration gradient using ATP
D Through the cell membrane by active transport

Circle your answer: A B C D

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6 Which blood vessel carries oxygenated blood away from the heart to the body?

A Vena cava
B Pulmonary artery
C Aorta
D Pulmonary vein

Circle your answer: A B C D

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7 A student investigates the effect of temperature on enzyme activity. At 80°C, the reaction rate falls to zero. The most likely reason is:

A The substrate has been fully used up

B The enzyme has been denatured
C The pH has become too alkaline
D The enzyme has lost its substrate

Circle your answer: A B C D

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8 Which of the following is a correct description of diffusion?

A Movement of molecules from low to high concentration requiring energy
B Movement of molecules from high to low concentration along a concentration gradient
C Movement of water molecules only through a partially permeable membrane
D Movement of molecules against a concentration gradient using ATP

Circle your answer: A B C D

---

9 The hormone insulin is produced by cells in the:

A Liver
B Kidney
C Pancreas
D Adrenal gland

Circle your answer: A B C D

---

10 Which of the following is the correct sequence in the digestive system?

A Oesophagus → Stomach → Small intestine → Large intestine
B Stomach → Oesophagus → Small intestine → Large intestine
C Mouth → Large intestine → Stomach → Small intestine
D Oesophagus → Large intestine → Stomach → Small intestine

Circle your answer: A B C D

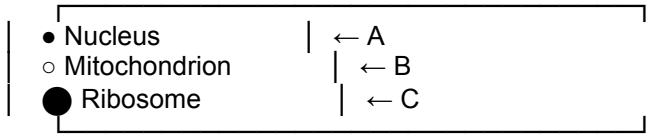
Section B: Short Answer and Structured Questions (100 marks)

11 Cells and Microscopy

Figure 1 shows a diagram of an animal cell as seen under a light microscope.

Figure 1

[Animal Cell Diagram]



(Arrow A = cell membrane, B = mitochondrion, C = nucleus — label positions marked)

(a) State the function of the structure labelled A.

(1 mark)

(b) Give TWO differences between a plant cell and the animal cell shown in Figure 1.

(2 marks)

1 .....

2 .....

(c) A student views the cell under a microscope. The image of the cell is 30 mm wide. The actual width of the cell is 0.03 mm. Calculate the magnification used.

Use the formula: magnification = image size ÷ actual size

(2 marks)

Working:

Magnification = ...

(d)* Explain how the structure of the cell membrane allows it to carry out its function.

(3 marks)

12 Enzymes

A student investigated the effect of pH on the activity of protease. The student measured the time taken for a piece of photographic film (coated with protein) to become clear at different pH values. Table 1 shows the results.

pH	Time taken (seconds)	Rate of reaction (1/time × 1000)
2	No reaction	—
4	560	1.8
6	210	4.8
8	95	10.5
10	240	4.2
12	No reaction	—

(a) State the optimum pH for this protease.

(1 mark)

Optimum pH = ...

(b) Calculate the rate of reaction at pH 4. Show your working.

(2 marks)

Working:

Rate = ...

(c) Describe and explain the effect of increasing pH above the optimum on enzyme activity.

(3 marks)

(d) The student did not control the temperature during the investigation. Explain why this could affect the reliability of the results.

(2 marks)

13 Human Nutrition and Digestion

Figure 2 shows the human digestive system.

Figure 2 — Human Digestive System

Mouth → Oesophagus → Stomach → Small intestine → Large intestine → Rectum → Anus

(Organs labelled A through G in the original diagram)

(a) Name the enzyme produced in the mouth and state the food molecule it digests.

(2 marks)

Enzyme: .....

Food molecule digested: .....

(b) Explain why the stomach produces hydrochloric acid.

(2 marks)

(c) The small intestine is adapted for the absorption of digested food. Describe THREE structural adaptations of the small intestine that increase the rate of absorption.

(3 marks)

(d) A patient is deficient in lipase. Explain the effect this would have on fat digestion and absorption.

(3 marks)

14 Respiration

A student measured the oxygen consumption of a small mammal at different temperatures. Table 2 shows the results.

Temperature (°C)	Oxygen consumed (cm^3 per hour)
5	18.2
10	14.1
15	10.3
20	8.8
25	8.7
30	8.9
35	9

(a) Describe the trend shown in Table 2 between 5°C and 25°C.

(2 marks)

(b) Suggest why the oxygen consumption is higher at 5°C than at 25°C for this mammal.

(2 marks)

(c) Write the word equation for anaerobic respiration in animals.

(1 mark)

..... → ..... + ...

(d) During intense exercise, a sprinter's muscles respire anaerobically. Explain why lactic acid builds up and how it is removed after exercise.

(4 marks)

15 Photosynthesis

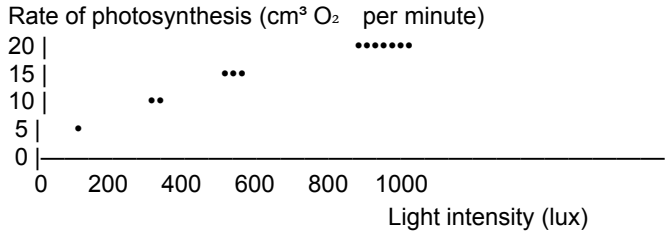
(a) State the word equation for photosynthesis.

(1 mark)

..... + ... → ..... + ...

(b) A student investigates the rate of photosynthesis in aquatic plants at different light intensities. Figure 3 shows a graph of the results.

Figure 3



(b)(i) Describe the relationship between light intensity and rate of photosynthesis shown in Figure 3.

(2 marks)

(b)(ii) Suggest why the rate of photosynthesis levels off at high light intensities.

(2 marks)

(c)* Explain the roles of chlorophyll, light and carbon dioxide in the process of photosynthesis.

(4 marks)

16 Transport in Plants

Figure 4 shows a cross-section through a leaf.

Figure 4 — Cross-section of a Leaf
Upper epidermis — Palisade mesophyll — Spongy mesophyll — Lower epidermis
Vascular bundle (X = xylem, Y = phloem) — Guard cells — Stomata

(a) Name the tissue labelled X and Y in the vascular bundle.

(2 marks)

X: .....

Y: .....

(b) Describe the process of transpiration.

(3 marks)

(c) A student measures the transpiration rate of a plant in different conditions. Table 3 shows the results.

Condition	Transpiration rate (cm^3/hour)
Still air, low humidity	1.2
Moving air, low humidity	2.8
Still air, high humidity	0.5
Moving air, high humidity	0.9

(c)(i) Identify the condition that gives the highest transpiration rate.

(1 mark)

.....

(c)(ii) Explain why moving air increases the rate of transpiration.

(2 marks)

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17 Inheritance

In pea plants, tall stem (T) is dominant over dwarf stem (t). Two heterozygous tall plants are crossed.

(a) State the genotypes of the two parent plants.

(1 mark)

Parent 1: ..... Parent 2: .....

(b) Complete the Punnett square below to show all possible offspring from this cross.

(2 marks)

	T	**t**
T		
t		

(c) State the expected phenotypic ratio of the offspring from this cross.

(1 mark)

.....

(d) A tall plant from the offspring is selected. Describe how you could determine whether this plant is homozygous or heterozygous tall.

(3 marks)

(e) Explain the difference between the terms 'genotype' and 'phenotype'.

(2 marks)

18 Ecology and the Environment

A woodland ecosystem contains the following organisms:

Oak tree → Caterpillar → Blue tit → Sparrowhawk

(a) Name the producer in this food chain.

(1 mark)

.....

(b) Using the food chain above, describe what would happen to the population of blue tits if the number of caterpillars decreased significantly. Give a reason for your answer.

(2 marks)

(c) The table below shows the energy flow through this ecosystem.

Trophic level	Energy (kJ/m²/year)
Oak tree (producer)	40000
Caterpillar (primary consumer)	4200
Blue tit (secondary consumer)	380
Sparrowhawk (tertiary consumer)	32

(c)(i) Calculate the percentage efficiency of energy transfer from oak tree to caterpillar.

(2 marks)

Working:

Efficiency = ... %

(c)(ii) Explain why energy is lost between each trophic level.

(3 marks)

19 Human Circulatory System

(a) State the main function of the circulatory system.

(1 mark)

.....

(b) Figure 5 shows the heart.

Figure 5 — The Human Heart

Right atrium — A
Left atrium — B
Right ventricle — C
Left ventricle — D
Atrioventricular valves — E

(b)(i) Name the blood vessel that carries blood FROM the lungs TO the heart.

(1 mark)

.....

(b)(ii) Explain why the left ventricle has a thicker muscular wall than the right ventricle.

(2 marks)

(c) Compare the structure and function of arteries and veins.

(4 marks)

(d)* Evaluate the risk factors associated with coronary heart disease. In your answer, include both lifestyle and non-modifiable factors.

(6 marks)

20 Practical Skills and Scientific Enquiry

A student wants to investigate the effect of light intensity on the rate of transpiration in a leafy shoot using a potometer.

(a) State the dependent variable in this investigation.

(1 mark)

.....

(b) Identify TWO variables that must be controlled to ensure a fair test.

(2 marks)

1 .....

2 .....

(c) Explain how the potometer is used to measure the rate of transpiration.

(3 marks)

(d) Table 4 shows the student's results.

Light intensity (lux)	Trial 1 (mm/min)	Trial 2 (mm/min)	Mean (mm/min)
100	1.2	1.4	
300	2.1	2.3	
500	3	2.9	
700	3.4	3.5	
900	3.4	3.6	

(d)(i) Complete the mean column for light intensities 300 and 500 lux.

(1 mark)

(d)(ii) One result appears anomalous. Identify this result and suggest a reason why it may have occurred.

(2 marks)

Anomalous result: .....

Reason: .....

(e) Suggest ONE way the student could improve the reliability of this investigation.

(1 mark)

Total for paper: 110 marks

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